Pranjal Samant

Robotics Software and Autonomy Engineer

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Professional Summary

Robotics Software & Autonomy Engineer with industrial R&D experience at The National Robotarium and a current focus on autonomous drones. Develops **C++** and **Python ROS** software for **cobots** and **AMRs**, covering motion planning, navigation, **SLAM**, computer vision and sensor integration. MSc Robotics graduate with peer-reviewed research and practical experience in simulation and embedded systems.

Technical Skills

Robot software: C++, Python, ROS, ROS 2, MoveIt, motion planning, navigation, SLAM
Perception and integration: Depth cameras, OpenCV, microcontrollers, C, ESP32, FreeRTOS
Tools and simulation: Linux, Git, Gazebo, PX4 SITL, MAVSDK, MAVLink, C#, MATLAB

Work Experience

Avionics Engineer (Robotics & Autonomy)

Aebocode Technologies | Panchkula, India — Oct 2026 - Present

- Develop software for autonomous drone systems in an avionics engineering role.
- Apply **ROS 2** in drone software development.
- Work with **visual SLAM** and **ArduPilot** for drone autonomy.

Co-Founder

Three Baltic Mountains — Aug 2026 - Present

- Co-founded a software venture building tools for productivity, focus and wellbeing.
- Contribute to **C#** and **WPF** development of **FocusSummit**, an offline Windows focus timer.
- Build focus and break workflows, timer state handling and local session-history features.

Independent Consultant

Self Employed | Remote, Edinburgh, UK — Jul 2026 - Sep 2026

- Advised early-stage ventures on robotics, control systems and automation.
- Translated product requirements into **system architectures** and **hardware specifications**.
- Guided **IoT** sensing and component selection for prototype development.
- Prepared procurement-ready **bills of materials** to support planning and investor discussions.

Robotics Software & Autonomy Engineer

The National Robotarium | Edinburgh, UK — Aug 2024 - Jun 2026

- Developed robotics software and autonomy features for **cobots** and autonomous mobile robots (**AMRs**) in industrial R&D.
- Designed and implemented **C++** and **Python ROS** nodes for robot-arm motion planning.
- Developed **ROS**-based mobile navigation, **SLAM** and computer-vision functionality.
- Integrated **depth cameras**, **microcontrollers** and sensor arrays with real-time robot control software.

Work Experience Continued

Software Engineer - Artificial Intelligence

CDAC Mohali | Mohali, India — Jan 2022 - Dec 2022

- Developed computer-vision and deep-learning models for automated medical image analysis.
- Built and trained neural networks in **Python** using **TensorFlow** and **Keras** for R&D prototypes.
- Created interactive web applications to demonstrate models and real-time predictions.

Selected Projects

Autonomous Solar-Farm Inspection and Drone Health Monitoring

Independent project | RideScan Integration Hackathon finalist | 2026 | Simulation
www.ridescan.ai/pranjal-samant

- Built a simulated Drone-in-a-Box system that flies autonomous solar-farm inspection missions and uses RideScan to monitor drone behaviour.
- Integrated **ROS 2**, PX4 SITL, Gazebo Harmonic and MAVSDK; logged flight telemetry at **50 Hz** and automated a **15-run** healthy baseline with CSV validation.
- Tested motor degradation against temporary wind disturbances; RideScan monitoring distinguished a persistent motor fault from a transient disturbance. Featured in RideScan's Builder Case Study.

AR and Voice Control for Robot Programming

MSc dissertation | Heriot-Watt University | 2024 | Gazebo simulation
github.com/pranjalsamant1/Robotic_Arm_Application

- Enabled hands-free control of a simulated robot arm through spoken commands on a RealWear Navigator 520 AR headset.
- Built a Unity Android app that sends commands over Wi-Fi to **ROS** via **ROS**Bridge/**ROS#**; MoveIt plans the arm movements and Gazebo simulates execution.
- Validated **60 of 60** commands across five sequences; measured **112.1 ms** mean response latency in 30 separate latency trials.

ESP32 Periodic Tasks and FreeRTOS

Embedded systems project | C/C++, ESP32, FreeRTOS
github.com/pranjalsamant1/FreeRTOS_Project

- Built an ESP32-based system that generates waveforms, measures signal frequency, filters analogue sensor inputs and logs readings over a serial interface.
- Reworked the sequential design with **FreeRTOS** tasks, **queues** and **mutexes**, coordinating button events and shared sensor data across concurrent tasks.

Education

MSc Robotics | Heriot-Watt University, United Kingdom | Jan 2023 - May 2024
Relevant study: motion planning, SLAM, human-robot interaction, embedded and autonomous systems.

Bachelor of Engineering | Electrical, Electronics and Communications Engineering
SRM Institute of Science and Technology, India | Apr 2018 - May 2022

Publication

Co-author: Vision-based manipulation of transparent plastic bags in industrial setups. Frontiers in Robotics and AI, 27 January 2025.
doi.org/10.3389/frobt.2025.1506290